

EVENT RECOMMENDATION SYSTEM

Connecting venue preferences with personalized local event discovery using AI.

Authors

Tavishi Kaushik, Simron Patel, Eden Growney



Affiliations

Northwestern University | MS in Artificial Intelligence

Abstract

This project proposes a hybrid recommendation system that links user venue preferences with local event discovery. Using the Yelp dataset, we apply collaborative filtering to learn user and venue representations from review data. Event data from the Ticketmaster API is embedded using a text embedding model and matched with venue embeddings through similarity search. Events associated with recommended venues are then mapped back to users to generate personalized event recommendations within their geographic area.

Problem Statement

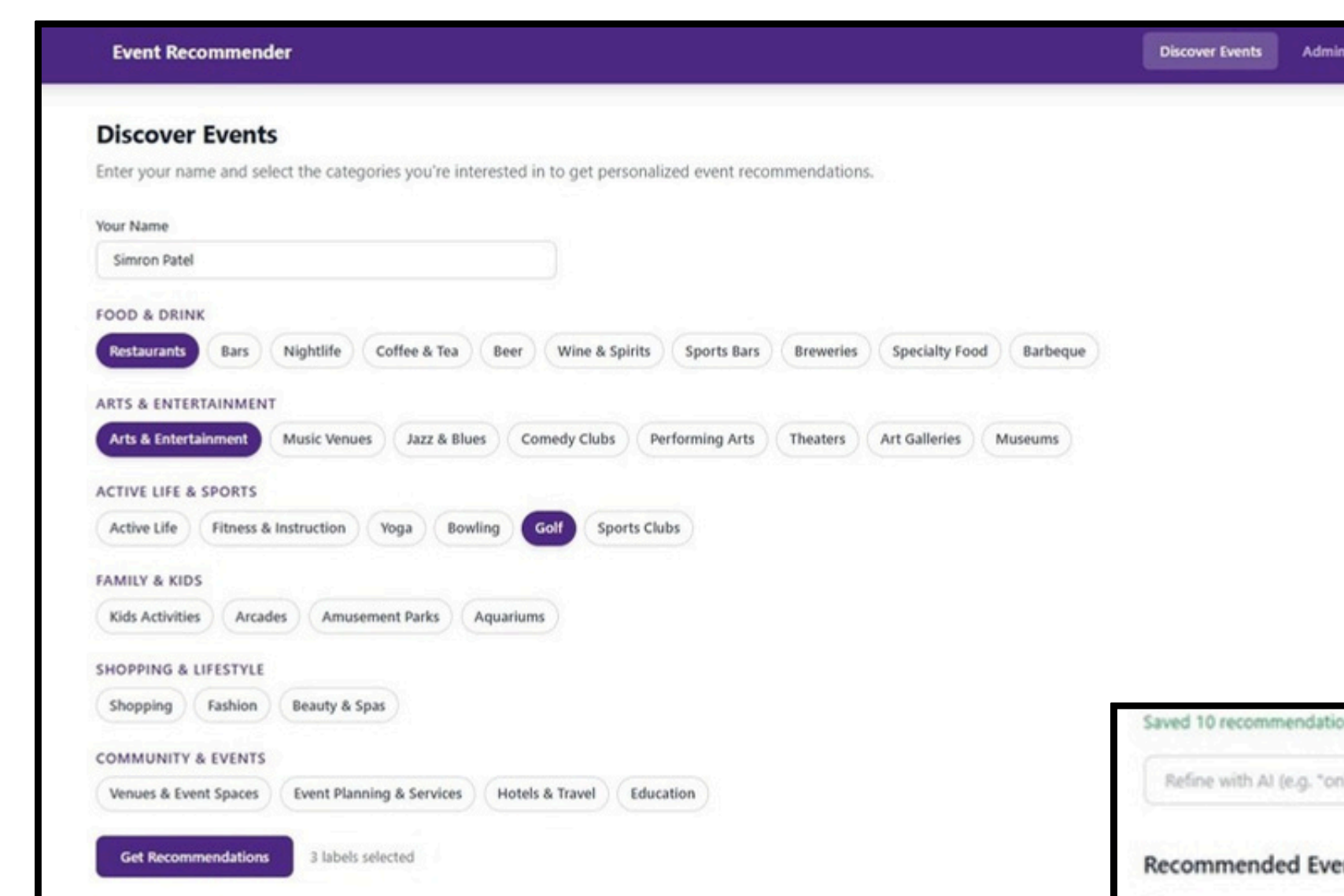
- Discovering relevant **local events** is difficult because many recommendation systems rely on simple filters such as **location** or **popularity** rather than **personal interests**.
- User behavior data** from platforms like **Yelp** contains valuable signals about individual preferences through **reviews** and **ratings of venues**.
- The **challenge** is to leverage this data to **infer user interests** and **connect them** to meaningful local events.
- This is difficult because the data sources and formats differ **between venue data** (e.g., Yelp) and event listings.

Solution

We created an Admin Interface that shows how the system makes recommendations. Instead of a normal user view that only shows one person's results, this view shows how preferences are learned across all users to understand transparency into how the system works

Step 1: Build User

Admin selects venue categories to simulate a user's interests based on places they might review or visit.

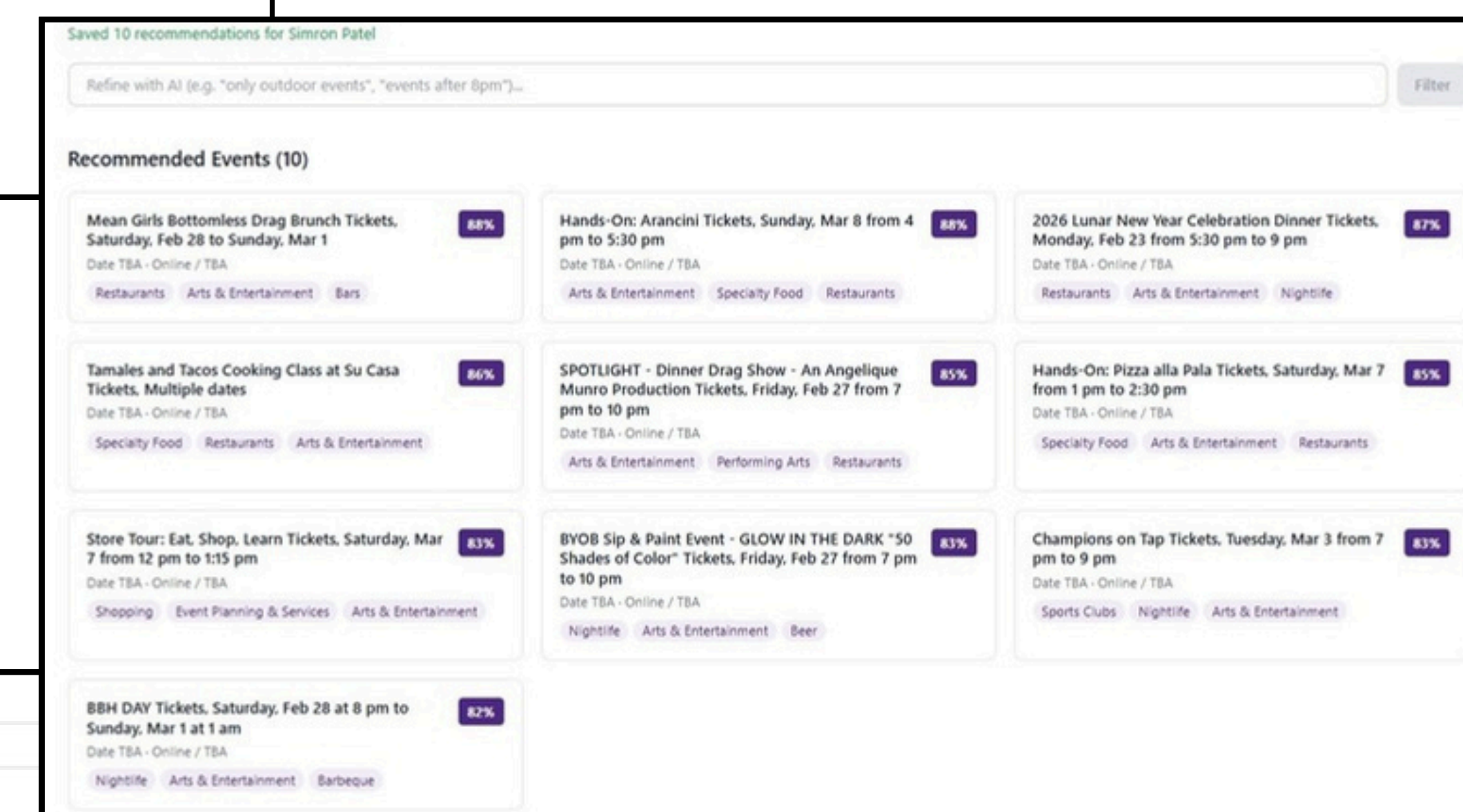


Step 2: Initial Recommendations

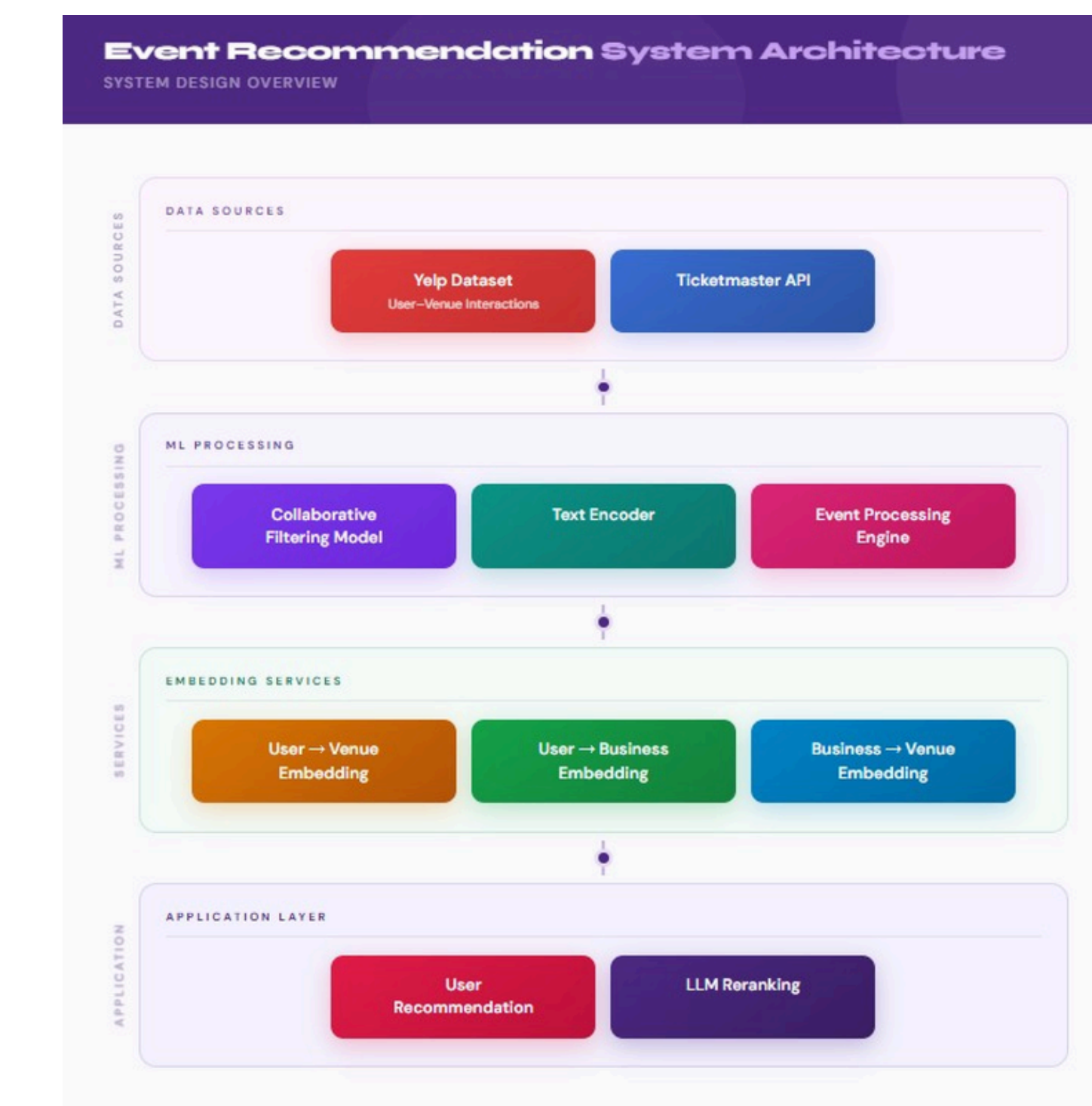
The system generates event suggestions by matching these interests with event categories.

Step 3: LLM Filtering

A language model refines the list by selecting key words that better align with the user's interests.



Architecture



Analysis

We evaluated our system by testing **how accurately it could identify the type of venue or activity** associated with events.

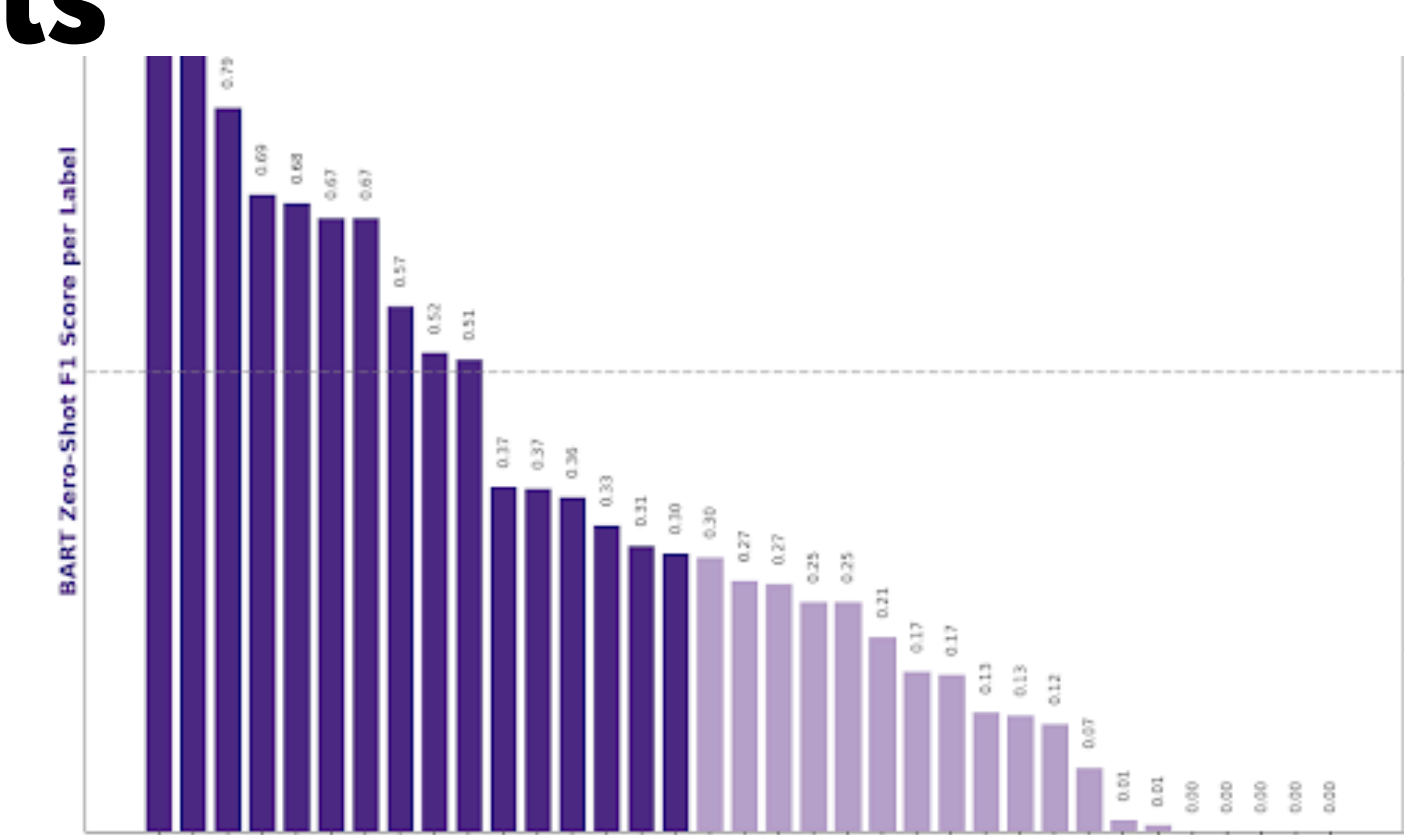
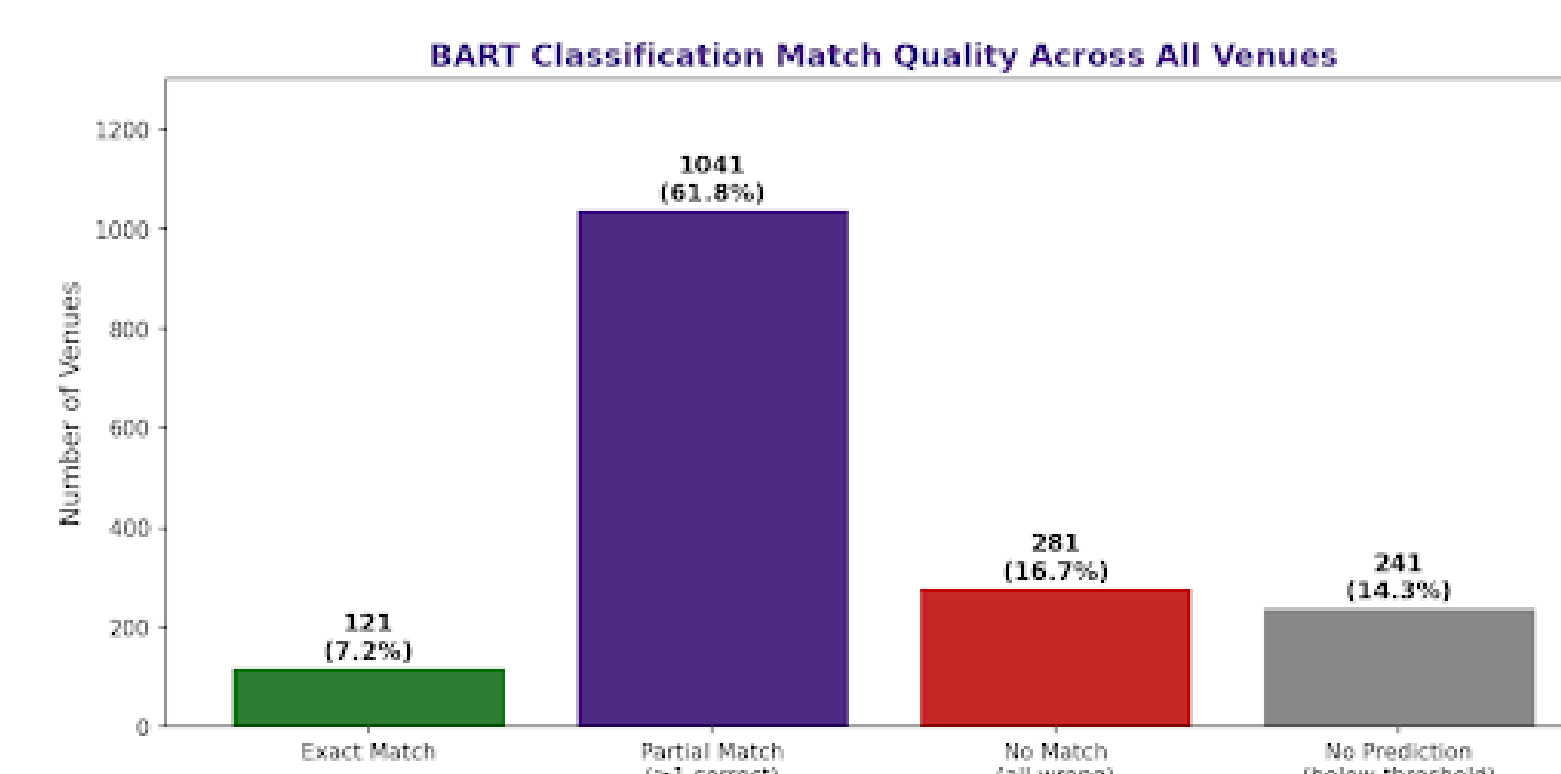
- Uses a **zero-shot approach** e.g predicts categories based on general language understanding rather than being trained on labeled examples.
- Predictions were compared to the actual venue categories to measure performance.

Results & Implications

- The system correctly identified at least one relevant category for **~62% of venues**.
- It performs **well for broad consumer categories** (e.g., restaurants, travel, nightlife).
- Performance is **weaker for more specific venue types** (e.g., performing arts or specialized venues).

Takeaway: The system works well for general event discovery, but improving recommendations for niche venues will require additional training data

Results



Key Sources

Yelp Open Dataset (<https://www.yelp.com/dataset>), Ticketmaster Discovery API (<https://developer.ticketmaster.com/>), Recommender Systems Handbook (Ricci, Rokach & Shapira (2015)), Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks (Reimers & Gurevych (2019)), Collaborative Filtering Recommender Systems (Koren, Bell & Volinsky (2009))